

MechIQ Flowchart

All these sensors will be combined in our unique plug-and-play non-invasive device.

SENSOR LAYER



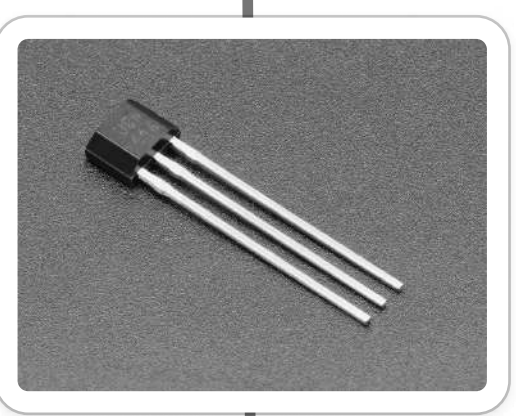
ADXL345 Vibration Sensor



SCT-013 Current Sensor



MLX90614 Temperature Sensor



A3144 RPM/Speed Sensor

BLE - Bluetooth Low Energy

EDGE PROCESSING LAYER

ESP32 S3
Microcontroller

FFT Processing

Feature
Extraction

Isolation
Forest

Anomaly Score
Detection

4G Cellular Connectivity

Local Storage - Micro SD card module. When cellular connection breaks, it will act as a local buffer.

CLOUD PROCESSING LAYER

InfluxDB/AWS

LSTM

FEEDBACK: Every alert resolved by the mechanic is classified (True Positive / False Alarm) via the app.

OUTPUT LAYER

WhatsApp Alert:
Gujarati/Hindi -
Owner + Mechanic

Web Dashboard -
Machine Health Scores

Mechanic Job Card
- Component +
Spare Part +
Supplier

Native language localized alerts eliminate the need for specialized plant IT literacy.

Identify Challenges	HMW Statements	Ideas from Outside	Reverse Brainstorming	Possible Constraints
What are the obstacles you want to move past?	Turn each challenge into a 'How Might We' question	Collect ideas from different industry perspectives	What could break our solution? How do we pre-solve it?	Limitations we face & our workarounds
Core pain points in the MSME maintenance gap 6.3 crore MSMEs lose Rs.15–40L/yr to unplanned breakdowns — zero affordable monitoring below Rs.5L exists today Every PdM system is English-only. Surat factory owners speak Gujarati — cannot use any existing platform Amazon Monitron (AWS) built this exact product, priced at Rs.350/month — shut it down Oct 2024. Generic approach failed at global scale. New imported machines (water jet, rapier) run at 600–1,000 RPM — bearing failure risk is 4–5× higher but OEM provides ZERO mechanical health monitoring	Design questions that frame our innovation How might we make PdM affordable for a 10-machine factory at Rs.199–299/machine/month — less than one mechanic call-out? How might we deliver machine failure alerts to a Gujarati-speaking owner with no laptop — in under 30 seconds via WhatsApp? How might we install on ANY rotating machine in under 30 minutes — no electrician, no drilling, no modification? How might we solve Amazon's three failures: slow setup → QR activation; noisy alerts → 7-layer filter; no context → spare part + supplier in alert? How might we turn the factory mechanic from a passive repairman into an active ML data contributor?	Cross-industry inspiration powering our USPs FROM TELECOM: WhatsApp has 500M Indian users. Use it as the delivery OS — zero new app, zero onboarding friction FROM SAAS/FINTECH: Convert Rs.5L capital cost to Rs.299/month opex. Zerodha proved Indian SMEs adopt when capex = zero FROM AUTOMOTIVE OBD: Pre-load 25 machine profiles (water jet, rapier, compressor) so system knows the machine before installation. Cuts cold start from 28 days to 3–5 days FROM FOOD DELIVERY (Zomato model): Alert fires → mechanic receives WhatsApp job card → taps Accept → arrives with pre-diagnosed part. Closes the loop Amazon never did FROM MICROFINANCE (Grameen/Amul): SGCCI partnership = 500 customers in one conversation. Cluster trust beats 1,000 cold calls	What situations could invalidate our solution? Too many false alerts → factory owners cancel in Month 2 and spread negative word-of-mouth across the cluster Consumer-grade hardware fails in 45° C dusty factory within 6 weeks Owner says: 'My mechanic knows by sound — why pay Rs.299/month?' Our pre-built solutions to each obstacle 7-LAYER FILTER: 28-day silent learning → 2-sensor agreement → 5-min persistence → 4 alert tiers → warm-up windows → mechanic feedback → 30-min trust recovery call + account credit IP54 enclosure + TVS surge protection + rubber vibration mounts + quarterly maintenance visit included in SaaS fee 'Your mechanic checks 1 machine. We can monitor all 50 simultaneously. Mechanic arrives with the correct part already identified.'	What limitations might we face? BIS CRS certification required before first commercial sale — 6–12 months, Rs.3L cost Cold start ML problem — no labeled failure data on Day 1 Working capital: Rs.54L needed for 6 pilot customers in Year 1 (after 50% advance payments) Student team — limited shipped hardware experience What are our workarounds? Submit BIS at STQC Ahmedabad Month 2 → certificate Month 5 → aligns with first paid sale Month 6. Train ML on CWRU + IMS open datasets (free). Results directly comparable to peer-reviewed literature. Apply for MSME Hackathon (Rs.15L) + NIDHI PRAYAS 2.0 (Rs.20L) + GUSEC simultaneously. Non-dilutive Rs.35L+ covers Year 1 fully. 50% advance contracts from customers.

TITLE: MechIQ Edge Node Architecture

REV: 1.0.0 | PATENT PENDING HARDWARE TOPOLOGY

[REDACTED IP: LDO & NOISE ISOLATION]

PREVENTS SMPS SWITCHING NOISE ON ADC/SPI

3.3V CLEAN

U2: ADXL345

HIGH-FREQ MEMS (SPI)

CS
SDI
SDO
SCLK

3200Hz DMA SPI BUS

U1: ESP32-S3-WROOM

EDGE INFERENCE & ML CORE

IO10 (CS)
IO11 (MOSI)
IO13 (MISO)
IO12 (SCK)

IO4 (ADC_CH3)

IO8 (SDA)
IO9 (SCL)

GND

[REDACTED IP BLOCK: ANALOG FRONT END]

PROPRIETARY BURDEN, BIAS & ANTI-ALIASING

AC IN

R_BURDEN

DC_OFFSET

AFE_OUT

SCT-013

U3: MLX90614

IR TEMP (I2C)

SDA
SCL